Seg2Reg-Net: An Explainable AI Analysis of Predictive Limitations in Cattle Weight Estimation

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Abstract

Accurate measurement of cattle body weight is critical for improving nutrition, health management, breeding strategies, and market value. Automated methods using computer vision and deep learning provide a promising non-contact alternative for weight estimation. However, these approaches face challenges because 2D images often fail to capture full body volume, key anatomical details, and variations caused by lighting, occlusion, or background clutter, which lead to inconsistent and inaccurate predictions. To address these challenges, we propose Seg2Reg-Net, an end-to-end framework that integrates an Attention U-Net-based segmentation module with a regression network. The proposed model achieves enhanced predictive accuracy, reducing RMSE from 53.36 with standard regression to 41.56. Pixel-level explainability techniques, including LIME and SHAP, are applied to systematically interpret model predictions. LIME highlights top positive contributors such as the mid-body and hindquarters, while SHAP reveals positive, negative, and near-zero effect regions, exposing spurious correlations and background influences. These findings contribute to the refinement of data preprocessing and model design, facilitating the practical application of accurate weight predictions in livestock management.

CCS Concepts

• **Computing methodologies → Explainable artificial intelligence**.

Keywords

Cattle weight estimation; explainable AI (XAI); computer vision; regression; segmentation

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1 Introduction

Accurate body weight estimation is a critical factor in livestock management. It directly guides livestock farmers to inform about feeding strategies, health monitoring, breeding decisions, and market valuation [9, 14]. Globally, more than one billion livestock are evaluated annually for productivity and commercial potential, but most conventional measurements remain labor-intensive, time-consuming, and often subjective [29, 36]. Traditional approaches, such as manual weighing on industrial scales or using body morphometrics, are difficult in practice because weighing requires large equipment and multiple workers. Manual body measurements also require close contact, making the process slow, costly, and inconsistent [5, 15].

Recent advances in non-contact weight estimation, using low-cost sensors and machine vision techniques, offer a promising alternative by automating morphometric data evaluation and reducing both handling stress and labor requirements [9]. By informing size measurements with additional animal characteristics such as age, sex, genotype, body condition, and volumetric parameters, predictive models can accurately estimate weight without direct physical contact [5]. Although traditional studies predominantly employ multiple linear regression for weight prediction, modern approaches increasingly combine computer vision with machine learning for greater accuracy and scalability [9]. These automated methods enable faster, less subjective, and more efficient livestock assessment, providing valuable insights for nutrition, health, and breeding management.

However, these modern automation strategies are not always reliable in livestock weight estimation due to several critical limitations. **Firstly**, failures in feature extraction remain a major cause of inaccuracy. In many cases, conventional algorithms do not capture essential body contours, volumetric information, and subtle morphometric descriptors when relying only on surface-level features [10, 19]. These limitations are further amplified when low-resolution or noisy images are used, and they lead to an incomplete representation of animal morphology and consequently reduce predictive accuracy. **Secondly**, the reliance on two-dimensional imaging techniques imposes fundamental constraints. Since 2D projections cannot fully represent the volumetric structure of an animal, important depth-related information such as body thickness or torso curvature is lost [15, 36].

As a result, weight estimates derived from 2D images often suffer from underestimation or overestimation errors, particularly when animals are not standing in standardized poses. Although 3D reconstruction and depth sensors have been proposed as alternatives, their high cost and complexity still limit adoption in large-scale farming environments. **Thirdly**, the variability of environmental and biological conditions poses significant challenges. Farm environments often introduce background noise such as overlapping animals, shadows, and cluttered surroundings, while biological differences such as breed-specific morphology, growth stages, and body condition reduce the generalizability of trained models [14, 33]. Models optimized for a specific breed or dataset often fail when applied to different herds or geographic contexts, which highlights the need for domain adaptation and larger, annotated datasets.

The limitations of conventional machine learning approaches underscore the pressing need for explainable AI (XAI) in livestock weight estimation. Traditional "black-box" models produce predictions without providing insight into the underlying decision-making process, which makes it difficult to identify why errors occur under certain conditions. XAI techniques, such as SHAP (SHapley Additive exPlanations) [16], LIME (Local Interpretable Model-agnostic Explanations) [22], Grad-CAM [24], and attention-based visualization methods [4], enable detailed interpretation of model outputs by highlighting which features, pixels, or regions most influence predictions. By applying these approaches, researchers can systematically trace failures to specific causes such as poor feature extraction, 2D imaging limitations, or environmental noise and subsequently refine preprocessing, segmentation, or training strategies. Moreover, integrating XAI not only improves model transparency and trustworthiness but also facilitates actionable insights for breeders and farm managers, bridging the gap between predictive accuracy and practical applicability [1, 3].

To address these limitations, we design a systematic study combining predictive modeling with explainable AI techniques. We begin by implementing a standard regression model to estimate cattle weight from images, establishing a baseline for comparison. Recognizing that errors often arise from incomplete or noisy representations of body morphology, we incorporate a segmentation stage to isolate the animal from the background. Building on this, we propose a unified architecture, Seg2Reg-Net, which integrates segmentation and regression in an end-to-end framework. Despite these improvements, accurate weight prediction remains challenging due to residual noise, intra-animal variation, and limitations of 2D imaging. To understand the sources of these errors, we apply pixel-level interpretability techniques, including LIME and SHAP, which reveal the specific regions and features contributing most to the predictions. By systematically analyzing these explanations, we identify patterns of model failure, such as underestimation in occluded or partially visible body parts, and overestimation in high-reflectance regions or complex backgrounds. This diagnostic insight informs further refinements in preprocessing, data augmentation, and model architecture to bridge the gap between high predictive accuracy and practical applicability. Our key contributions are given below:

- We propose Seg2Reg-Net as an end-to-end architecture that integrates segmentation and regression to improve the accuracy of cattle weight estimation from 2D images. For the segmentation module, an Attention U-Net is employed to generate precise segmentation masks, which are subsequently processed through residual blocks in the regression stage to enhance predictive performance.
- Pixel-level explainability is employed using LIME and SHAP to systematically interpret model predictions. The analysis reveals that the model frequently attributes importance to anatomically irrelevant regions and background artifacts, indicating reliance on spurious correlations. These findings underscore the need for improved preprocessing and network design to ensure biologically meaningful feature attribution.

2 Related Work

The estimation of weight from images has received considerable attention across diverse domains such as agriculture, healthcare, and dietary assessment. Recent advancements in deep learning and computer vision have enabled more accurate, non-invasive methods for weight prediction. In dietary assessment, estimating the weight of food items from images is crucial for nutritional analysis. Wimalasiri and Sahoo [30] proposed a vision-based approach using 2D images, combining Faster R-CNN for food detection with MobileNetV3 for weight estimation. Similarly, Pham et al. [21] extracted image features and trained them using boosting regression algorithms on a dataset of 23,052 annotated food images. More recent approaches leverage 3D reconstruction to enhance accuracy. Ma et al. [18] introduced MFP3D, which generates 3D point clouds from monocular images to estimate food portions, while Shao et al. [25] reconstructed 3D food shapes for energy estimation. Han et al. [11] further enhanced performance by fusing depth prediction with RGB data for accurate nutrition estimation.

Estimating human body weight from images can support patient monitoring and treatment planning. Shu et al. [26] developed a system using a single image to estimate height and weight of infants through MobileNet-based models. Debnath et al. [8] introduced the Celeb-FBI dataset, applying deep learning models such as ResNet-50 and VGG-16 for full-body weight estimation. Lee et al. [13] proposed a CNN-based method to estimate body weight from chest and abdominal CT scout images.

In agriculture and livestock management, weight estimation aids in monitoring and optimizing animal growth. He et al. [12] utilized RGB-D images with LiteHRNet for sheep weight prediction, while

Zhang et al. [35] developed a machine learning approach for real-time pig weight estimation from RGB images. Point cloud-based approaches also show promise; Zhang et al. [34] demonstrated accurate finishing pig weight prediction even under slight posture variations. Multimodal methods that combine RGB, depth, and 3D point clouds have been applied to duck weight estimation [32].

Pressure image-based weight estimation has emerged as a non-invasive and privacy-preserving approach. Wu et al. [31] introduced MassNet, a dual-branch network that extracts deep and pose features from a single pressure image, outperforming existing methods. Smith et al. [27] applied YOLOv8 for pressure injury assessment in clinical scenarios.

3 Methods

The proposed methodology summarizes the complete operational workflow of the Seg2Reg-Net framework, which is designed for estimating cattle weight as shown in Figure 1. The process begins with data collection, followed by some preprocessing steps, and then the model moves through two key stages. In Stage one, an Attention U-Net segmentation network is used to provide accurate and reliable cattle body masks while minimizing background interference. These masks are then applied to the images to focus on the relevant areas. In Stage two, masked images are fed into a residual CNN-based regression network, which outputs the cattle weights. Appropriate loss functions were incorporated in model training: the binary cross-entropy (BCE) with Dice loss for segmentation and the mean squared error (MSE) for regression are used. Besides, explainability methods: LIME and SHAP were incorporated to visualize the images, which regions were important to the prediction, and improve the interpretability. The approach as a whole integrates segmentation, regression, performance evaluation, and explainability into an end-to-end pipeline for accurate cattle weight prediction.

3.1 Dataset Preprocessing

We denote our dataset as $\mathcal{D} = \{(x_i, y_i, v_i, g_i)\}_{i=1}^{N}$, where $x_i \in \mathbb{R}^{H\times W\times 3}$ represents the input RGB image, $y_i \in \mathbb{R}$ is the corresponding cattle weight in kilograms, $v_i \in \mathcal{V}$ is the categorical variable indicating the view (side or rear), and $g_i \in \mathcal{G}$ represents the gender annotation. Here, N is the total number of samples in the dataset. We first standardize the input images by resizing them to a fixed resolution, $x_i' = \text{Resize}(x_i, (224, 224))$, where x_i' is the resized image of dimension $224 \times 224 \times 3$. We then normalize their pixel intensity values into the unit range,

$$\tilde{x}_i = \frac{x_i'}{255.0}, \quad \tilde{x}_i \in [0,1]^{224\times224\times3},$$

where $\tilde{x}_i$ represents the normalized input image. We handle the annotation masks $m_i \in \mathbb{R}^{224\times224\times3}$ by extracting the green channel, $m_i^{(g)} = m_i[:,:,1]$, where $m_i^{(g)}$ represents the green channel of the mask indicating cattle regions. We generate a binary mask through thresholding,

$$\hat{m}_i(u,v) = \begin{cases} 1, & m_i^{(g)}(u,v) > 0.5, \\ 0, & \text{otherwise}, \end{cases}$$

where $\hat{m}_i(u,v)$ is the binary mask at pixel (u,v). Finally, we construct two task-specific datasets. For segmentation, we define $\mathcal{D}_{seg} = \{(\tilde{x}_i, \hat{m}_i)\}_{i=1}^{N}$, where $\mathcal{D}_{seg}$ represents the dataset for training the segmentation model. For regression, we define $\mathcal{D}_{reg} = \{(\hat{m}_i, y_i)\}_{i=1}^{N}$, where $\mathcal{D}_{reg}$ represents the dataset for training the regression model to predict cattle weight from the segmented masks.

3.2 Segmentation Network

In the segmentation stage, we employ an Attention U-Net to extract cattle regions from preprocessed images. Let $\tilde{x}_i \in [0,1]^{224\times224\times3}$ denote the input image after resizing and normalization. The network predicts a segmentation mask $\hat{m}_i = f_\theta^{seg}(\tilde{x}_i), \quad \hat{m}_i \in [0,1]^{224\times224\times1}$, where $\hat{m}_i$ is the predicted probability mask for the cattle region and f_θ^{seg} represents the Attention U-Net parameterized by θ.

To improve focus on relevant features, we incorporate attention gates in the skip connections. Let $x \in \mathbb{R}^{h\times w\times C_x}$ denote the encoder feature map, and $g \in \mathbb{R}^{h\times w\times C_g}$ denote the gating signal from the decoder. The attention coefficients are computed as $\alpha = \sigma\left(W_x x + W_g g + b\right)$, where $W_x \in \mathbb{R}^{C_{int}\times C_x}$ and $W_g \in \mathbb{R}^{C_{int}\times C_g}$ are learnable linear weights, $b \in \mathbb{R}^{C_{int}}$ is a bias term, C_{int} is the number of intermediate channels, and $\sigma(\cdot)$ is the sigmoid activation function. The output of the gated skip connection is then $\tilde{x} = \alpha \odot x$, where $\odot$ denotes element-wise multiplication, allowing the network to selectively emphasize informative spatial regions while suppressing irrelevant background.

We train the network using a combination of binary cross-entropy and Dice losses to handle class imbalance between cattle and background pixels. Let $m \in \{0,1\}^{H\times W}$ denote the ground-truth binary mask and $\hat{m} \in [0,1]^{H\times W}$ denote the predicted mask. The binary cross-entropy loss is defined as

$$\mathcal{L}_{BCE} = -\frac{1}{HW}\sum_{u=1}^{H}\sum_{v=1}^{W}\Big[m(u,v)\log\hat{m}(u,v) + (1-m(u,v))\log(1-\hat{m}(u,v))\Big],$$

where H and W are the height and width of the mask, and $m(u,v)$, $\hat{m}(u,v)$ are the ground-truth and predicted values at pixel (u,v).

To further encourage spatial overlap between predicted and true masks, we define the Dice coefficient as

$$\text{Dice}(m,\hat{m}) = \frac{2\cdot|m\cap\hat{m}|}{|m|+|\hat{m}|} = \frac{2\sum_{u,v} m(u,v)\hat{m}(u,v)}{\sum_{u,v} m(u,v) + \sum_{u,v}\hat{m}(u,v)},$$

where $|m|$ and $|\hat{m}|$ denote the total number of foreground pixels in the ground-truth and predicted masks, respectively. The corresponding Dice loss is $\mathcal{L}_{Dice} = 1 - \text{Dice}(m,\hat{m})$, which penalizes poor overlap between prediction and ground-truth. Finally, we define the total segmentation loss as $\mathcal{L}_{seg} = \mathcal{L}_{BCE} + \lambda\cdot\mathcal{L}_{Dice}$, where $\lambda \in \mathbb{R}^{+}$ is a weighting hyperparameter controlling the contribution of the Dice loss. Minimizing $\mathcal{L}_{seg}$ encourages the model to produce accurate and well-aligned masks, effectively delineating the cattle region in each input image.

3.3 Regression Network

Once the segmentation masks $\hat{m}_i$ are generated, we use them as structured inputs to predict cattle weights. Let $\hat{m}_i \in [0,1]^{224\times224\times1}$ denote the predicted mask for the i-th image. We train a regression

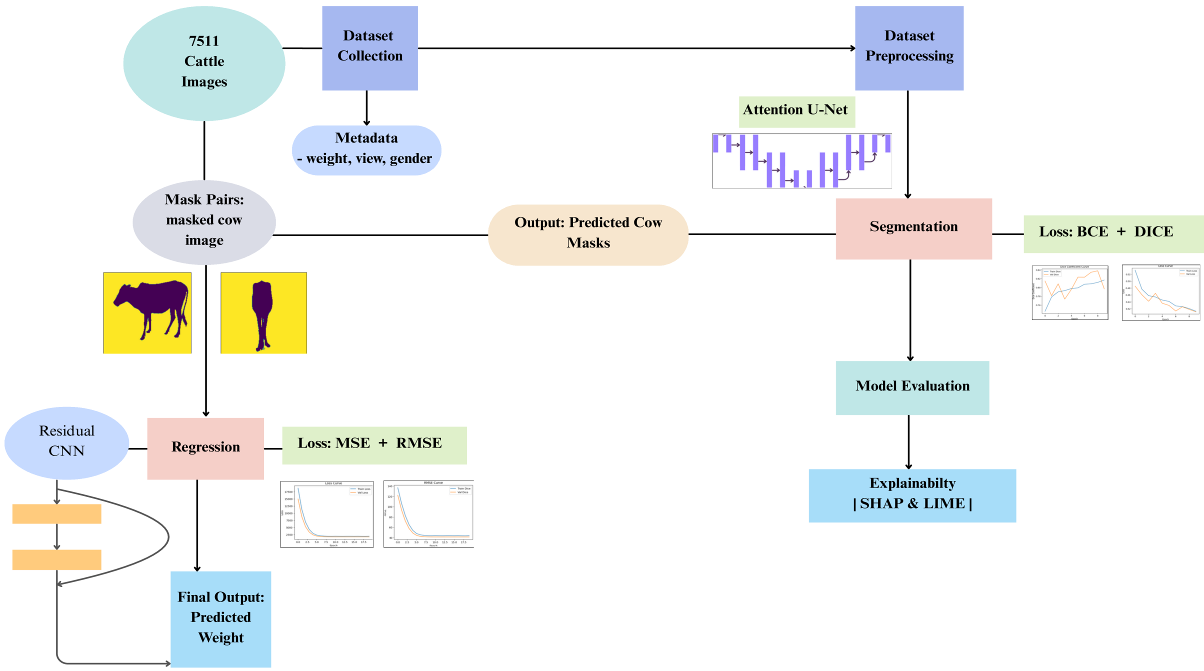


Figure 1: Methodology pipeline of Seg2Reg-Net for cattle weight prediction.

model parameterized by ϕ, denoted as f_{ϕ}^{reg}, to predict the corresponding weight $\hat{y}_i \in \mathbb{R}$: $\hat{y}_i = f_{\phi}^{reg}(\hat{m}_i), \quad \hat{y}_i \in \mathbb{R}, \quad i = 1, \ldots, N$ where N is the total number of samples in the regression dataset $\mathcal{D}_{reg}$.

Our regression network uses *residual blocks* to facilitate deeper learning and gradient flow. Each residual block takes an input feature map $z \in \mathbb{R}^{h \times w \times c}$ and applies two convolutional transformations with batch normalization (BN) and a non-linear activation function σ (here we use tanh) to produce an intermediate feature map z': $z' = \sigma\Big(\mathrm{BN}(W_2 * \sigma(\mathrm{BN}(W_1 * z)))\Big)$, where $*$ denotes the convolution operation, and W_1, W_2 are learnable convolutional weight tensors. To maintain the residual identity, we define a projection $\mathcal{P}(\cdot)$, which is either the identity mapping or a 1×1 convolution to match the channel dimensions. The output of the residual block is then given by: $z_{out} = z' + \mathcal{P}(z)$, ensuring that the block learns the residual mapping $\mathcal{F}(z) = z'$, which improves training stability and allows deeper network architectures.

To train the regression network, we minimize the *Mean Squared Error (MSE)* between the ground-truth weight $y_i \in \mathbb{R}$ and the predicted weight $\hat{y}_i$: $\mathcal{L}_{reg} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2$, where the sum is taken over all samples $i = 1, \ldots, N$. Minimizing $\mathcal{L}_{reg}$ encourages the model to produce predictions that are close to the true weights in a least-squares sense.

3.4 Explainability Analysis

To interpret the regression model's predictions, we employ two model-agnostic explainability techniques: SHAP and LIME. Let the regression model be denoted by f_{ϕ}^{reg}, taking an input mask $\hat{m}_i$ and

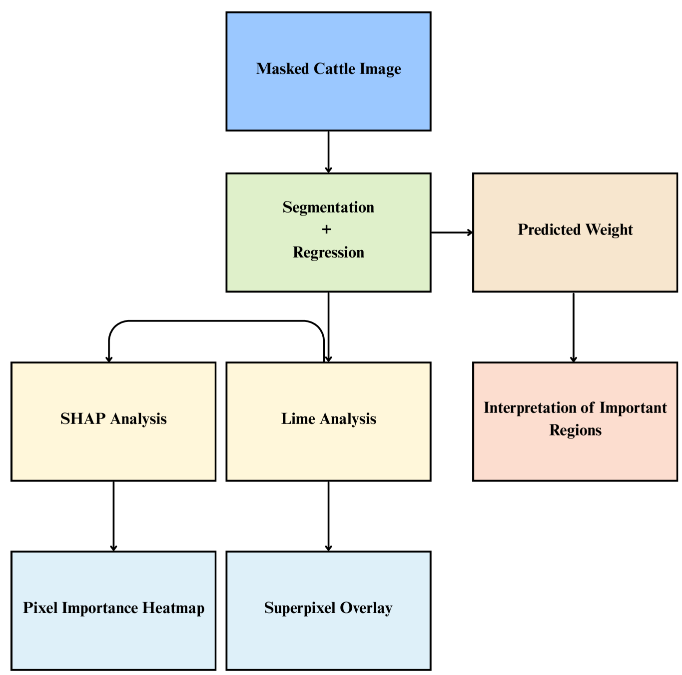


Figure 2: Explainability Analysis Pipeline of Seg2Reg-Net Using SHAP and LIME.

producing a predicted weight $\hat{y}_i$: $\hat{y}_i = f_{\phi}^{reg}(\hat{m}_i), \quad \hat{y}_i \in \mathbb{R}, \quad i = 1, \ldots, N$. The detailed overview is shown in Figure 2.

LIME Explanations. For localized perturbation analysis, we apply LIME. We define a segmentation function $\mathcal{S}(\cdot)$ that partitions the input crop into superpixels, collapsing all background pixels outside the cow mask into a single segment: $s_{i,k} = \mathcal{S}(x_i \odot \hat{m}_i), \quad k = 1, \ldots, K$, where K is the total number of superpixels and x_i is the cropped RGB image. We then generate N_s perturbed samples by randomly masking superpixels: $\tilde{x}_i^{(j)} = x_i \odot (\hat{m}_i \odot M^{(j)}), \quad j = 1, \ldots, N_s$, where $M^{(j)}$ is a binary mask selecting which superpixels remain visible. We pass each perturbed sample through the regression model: $\hat{y}_i^{(j)} = f_\phi^{reg}(\tilde{x}_i^{(j)})$ and map the continuous prediction $\hat{y}_i^{(j)}$ to a pseudo-probability for a two-class classification (light vs heavy cattle):

$$p_{i,heavy}^{(j)} = \frac{1}{1 + \exp(-(\hat{y}_i^{(j)} - \mu)/\sigma)}, \quad p_{i,light}^{(j)} = 1 - p_{i,heavy}^{(j)},$$

where μ and σ are the median and standard deviation of training weights. Finally, LIME fits a linear surrogate model on the perturbed samples $(\tilde{x}_i^{(j)}, p_i^{(j)})$ to assign importance weights β_k to each superpixel: $\hat{y}_i \approx \sum_{k=1}^{K} \beta_k \, s_{i,k}$, highlighting which cow regions in the input image were most influential for the predicted weight. We visualize these contributions as colored superpixels over the original image.

SHAP Explanations. We define a set of background samples $\mathcal{B} = \{\hat{m}_j\}_{j=1}^{B}$ to estimate the expected model output. The SHAP values $\phi_{i,u,v,c}$ for the i-th input, pixel (u, v), and channel c quantify the contribution of each feature to the prediction:

$$\hat{y}_i = \mathbb{E}[f_\phi^{reg}(\hat{m})] + \sum_{u=1}^{H} \sum_{v=1}^{W} \sum_{c=1}^{C} \phi_{i,u,v,c},$$

where $\mathbb{E}[f_\phi^{reg}(\hat{m})]$ is the expected model output over the background set. We attempt the GradientExplainer, which computes:

$$\phi_{i,u,v,c} \approx (\hat{m}_i - \bar{m}) \odot \frac{\partial f_\phi^{reg}}{\partial \hat{m}_i},$$

where $\bar{m}$ is the mean of background masks and $\odot$ denotes element-wise multiplication. If the gradient-based approximation fails, we use a generic SHAP explainer with an image masker $M(\cdot)$, which perturbs regions of the input while respecting the cow mask: $\phi_i = \mathrm{SHAP}(f_\phi^{reg}, M, \hat{m}_i, \mathcal{B})$, which produces pixel-level importance maps that indicate which regions of the mask contributed positively or negatively to the predicted weight.

4 Experiment and Evaluation

4.1 Dataset Description

The dataset is collected by Acme AI, Bhalo Social Enterprises, and mPower Social Enterprises as part of the *Smart Farming Innovations for Small-Scale Producers Global Grand Challenge* organized by the Bill & Melinda Gates Foundation [2], publicly available on kaggle. The dataset contains 7,513 images with corresponding semantic segmentation masks and 11,938 keypoint annotations. B2 and B4 include both side and rear views, while B3 contains only side views. Segmentation covers four classes-sticker, cattle, background, and ground-and key points mark body landmarks on side and rear images. Images are provided in standard formats such as `.jpg`, `.jpeg`, and `.png`. The dataset is split into 4,506 training, 1,502 validation, and 1,503 test images for model training and validation.

4.2 Experimental Setup

Evaluation Metrics. We evaluate the models using standard regression metrics including Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and the coefficient of determination (R^2) [6]. Additionally, we interpret the models using explainable AI techniques, namely LIME (Local Interpretable Model-agnostic Explanations) [23] and SHAP (SHapley Additive exPlanations) [17], to understand the contribution of individual features and image regions to the predictions.

Implementations and Model Details. From the segmentation network, we use an Attention U-Net architecture. The encoder contains three convolutional blocks with 32, 64, and 128 filters, each with two 3×3 convolutional layers with ReLU activation and same padding, followed by a 2×2 max-pooling layer. The decoder mirrors the encoder with upsampling layers, incorporating attention gates with 64 and 32 filters for the first and second upsampling stages. The final layer applies a 1×1 convolution with sigmoid activation to produce a single-channel binary mask. The model is optimized using the Adam optimizer with a default learning rate.

From the regression phase, we design a residual network to predict weight from segmented masks. The input layer accepts 224×224×1 images, followed by a 3×3 convolution with 32 filters, batch normalization, tanh activation, and a 2×2 max-pooling layer. Three residual blocks follow, each with two 3×3 convolutions having 64, 128, and 256 filters, paired with batch normalization and ReLU activation. 1×1 convolutions match filter dimensions for shortcut connections. Max-pooling layers are applied after the first two residual blocks, and dropout layers with rates of 0.2 and 0.5 are included after the first and third residual blocks. A global average pooling layer reduces spatial dimensions, followed by a dense layer with 256 units and tanh activation, and a final dense layer with a single unit for weight prediction. Both models use a batch size of 32 with TensorFlow prefetching data pipeline. The segmentation model trains for 10 epochs, and the regression model for 20 epochs.

4.3 Results and Discussion

We evaluate four models for cow weight prediction as shown in Table 1. The simple regression model performs poorly (MSE = 2847.26, RMSE = 53.36, MAE = 43.14, R^2 = -0.878). MobileNetV2 shows improved accuracy (MSE = 2013.66, RMSE = 44.87, MAE = 22.39, R^2 = 0.230). ResNet50 achieves the lowest MSE (1709.97). Our Seg2Reg-Net first applies segmentation and then regression, showing competitive performance (MSE = 1727.53, RMSE = 41.56, MAE = 31.65, $R^2 \approx 0$). The results indicate that our model struggles to produce accurate weight predictions from the images in all cases. To investigate these shortcomings, we perform pixel-level analyses using LIME and SHAP, highlighting the pixels that most strongly influence the predictions. Many prior studies have employed SHAP and LIME for image-based interpretation [7, 20, 28]. The SHAP and LIME studies consistently identify anatomically significant regions, including the mid-flank, dorsal ridge, and hindquarters, as the principal determinants of the model's predictions. This matches well with traditional body-measurement practices, suggesting that the model is indeed focusing on biologically relevant cues. At the same time, variations in the secondary regions indicate that some predictions are still being affected by residual noise or subtle shape distortions. These inconsistencies highlight the need for more robust data augmentation strategies and the incorporation of multiple views to enhance the model's stability and performance. This enables us to assess whether the extracted features are sufficiently informative for reliable weight estimation. This section presents a detailed analysis of each model below.

Table 1: Comparison of weight prediction performance across different models.

Model	MSE	RMSE	MAE	R^2 Score
Regression	2847.26	53.36	43.14	-0.878
MobileNetV2	2013.66	44.87	22.39	0.230
ResNet50	1709.97	–	31.54	–
Seg2Reg-Net (ours)	1727.53	41.56	31.65	-0.002

Result of Regression. We first establish the baseline regression model, which exhibits limited predictive accuracy, yielding an RMSE of 53.36, and a negative R^2 score of −0.878. This outcome indicates poor generalization and a failure to capture the underlying relationships between image features and cattle weight.

From Figure 3 (a–c), the LIME analysis shows that the mid-flank region (corresponding to the heart-girth zone) emerges as the most critical predictor, consistent with traditional manual weight estimation formulas. Additional focus is observed on the head and neck, which reflect frame size; the legs and hindquarters, which suggest skeletal scaling; and the mid-body and back, which capture musculature and fat deposits. Despite these localized attentions, the regression model often misinterprets contour variations and produces unstable contributions across regions. In particular, positive polarity (yellow outlines) highlights regions that inflate predictions when emphasized, while adjacent areas contribute inconsistently. This led to noisy and unreliable weight estimates, as even background regions are also attended to in the explanation maps, introducing noise.

From images (a)–(c) in Figure 4, the model primarily relies on cues related to body volume, musculature, and skeletal proportions, with the strongest positive contributions concentrated in the mid-flank and upper back. Additional but weaker positive effects are observed in the hindquarters, where hip width and muscle tone provide secondary evidence of larger body size. In contrast, the head and lower legs exert a negative influence, as narrower muzzles, smaller horns, or thinner limbs reduce the predicted weight and may serve as proxies for frame size. Minor negative contributions also arise from the background, reflecting limited sensitivity to environmental artifacts, whereas regions such as the belly and foreground appear largely neutral, which suggests either redundancy with flank information or irrelevance to weight estimation.

Result of Segmentation. We generate segmentation masks to reduce noise in the regression model, as illustrated in Figure 5. We

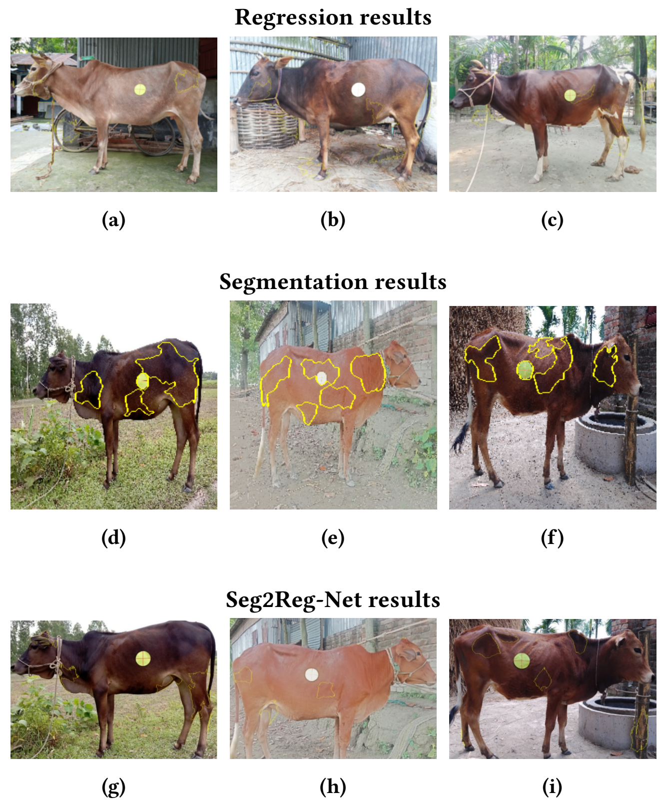


Figure 3: Comparison of model explanations using LIME for three approaches: Regression (a–c), Segmentation (d–f), and Seg2Reg-Net (g–i). The yellow-highlighted regions denote top positive contributors, termed *superpixels*. It indicates areas most influential in weight prediction.

achieve a mean Dice coefficient of 0.8326 on the test set as shown in Figure 6, indicating accurate delineation of the cow from the background. The attention maps from Figure 3 (d–f) show that the model predominantly focuses on anatomically relevant regions. The mid-flank, marked by a central yellow circle with crosshair, emerges as the most critical region, similar to a regression model. We also observe attention to the head and neck, which reflects body length and frame size, as well as to the legs and hindquarters, providing cues for skeletal structure and height. Contours along the mid-body and back capture musculature and fat deposits, indicating sensitivity to key features associated with body condition and weight. The yellow outlines highlight regions that positively contribute to the predicted weight, with the central marker serving as the primary focal point.

In Image (d) from Figure 4, positive contributions are concentrated on the torso and head, indicating that the model relies strongly on these body regions to confirm cow presence. Negative contributions appear around the legs, reflecting lower confidence due to their thin structure and potential confusion with the background. In Image (e), positive contributions extend uniformly across the entire body, suggesting high confidence in the full silhouette of the cow. Minimal negative contributions indicate that background regions are largely ignored, and a few pixels reduce the prediction of the model. In Image (f), the shoulders and mid-section show strong positive contributions, while yellow areas near the legs and tail indicate lower certainty in thin or occluded regions. Overall, the model consistently treats the central body mass as the most reliable feature for foreground segmentation, with peripheral regions contributing less to the predicted mask.

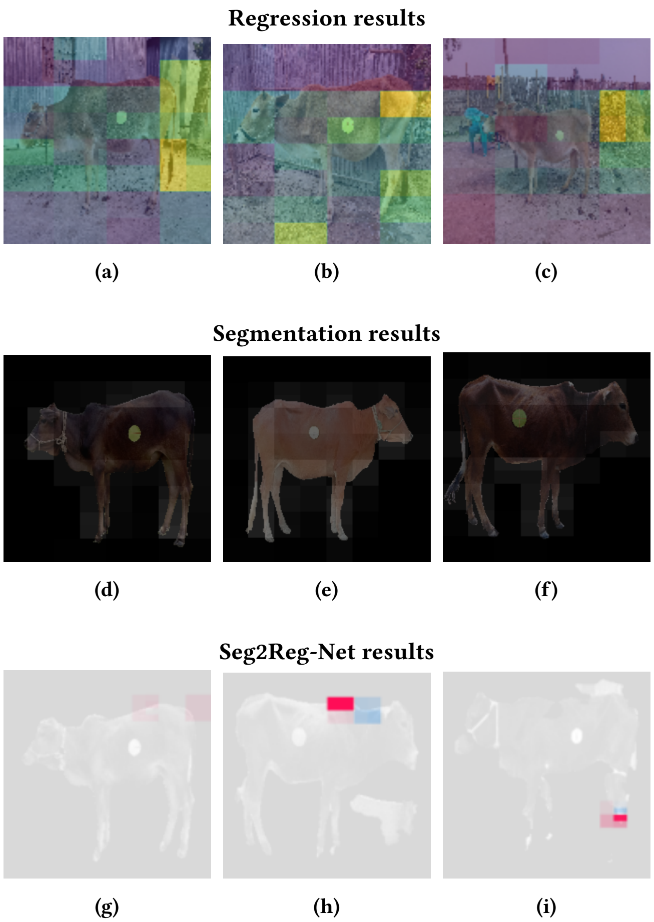


Figure 4: SHAP explanations for three models: Regression (a–c), Segmentation (d–f), and Seg2Reg-Net (g–i). Regression: Yellow/red → positive contributors, blue/purple → negative contributors; intensity shows effect strength. Segmentation: Pink → positive contributors, yellow → negative, gray → neutral. Seg2Reg-Net: Pink/red → positive, blue → negative, gray → near-zero effect.

Result of Seg2Reg-Net. By integrating segmentation with regression in Seg2Reg-Net, the model outperforms the baseline regression, achieving a reduction of 11.80 in RMSE and 11.49 in MAE. Still it indicates notable deviations from the actual weights. From images (g), (h), (i) in Figure 3, we show that the model predominantly focuses on the mid-body, including the flank and belly. It also attends to secondary regions such as the head, neck, legs, and hindquarters. The central marker in the LIME overlays consistently highlights the flank, aligning closely with traditional measurement sites. Interestingly, the model tends to overemphasize regions commonly used for manual measurements-head, limbs, and mid-body-treating them as dominant contributors to weight prediction.

From SHAP visualization images (g), (h), (i) in Figure 4, we observe that Seg2Reg-Net focuses on localized anatomical regions when estimating cattle weight. The upper back or shoulder region (red) contributes positively to the predictions, which indicates that the model interprets pronounced musculature and dorsal volume as signals of higher weight. Conversely, the adjacent neck-forequarter region (blue) contributes negatively, which illustrates that concavity or leanness in this area reduces the predicted weight. Most other regions, including the head, legs, belly, tail, and background exhibit near-zero influence, which demonstrate that the model selectively attends to core body mass indicators.

However, the 2D input images inherently limit volumetric assessment and regions that significantly influence weight variations. Our model fails to extract features from crucial parts like subcutaneous fat deposits around the shoulder, pelvis, and abdominal contour. Consequently, while the model assumes that commonly measured parts are the primary determinants of weight, subtle anatomical boundaries and edges that contribute to real-world weight differences are not fully captured. This limitation partly explains the observed prediction errors and highlights the challenges of estimating weight solely from 2D side- and rear-view images.

We also present scatter plots comparing predicted versus true weights for a special test set under ReLU and Tanh activation functions, as shown in Figure 7. With ReLU, the predictions are restricted to a narrow range around 156–170, failing to capture the variability of true weights and converging toward an average value. With Tanh, the outputs collapse to very small values (0.039–0.143), producing a severe mismatch in scale and no meaningful correlation with ground truth. ReLU gives values closer to the true range, but both activations fail to predict accurately.

5 Conclusion

Accurate estimation of cattle body weight remains challenging due to limitations in 2D imaging, incomplete feature extraction, and variability in environmental and biological conditions. This study proposes Seg2Reg-Net, an end-to-end framework integrating Attention U-Net-based segmentation with a regression network, and uses pixel-level explainability techniques such as LIME and SHAP to analyze model behavior. The analyses reveal that the model often relies on spurious correlations and background artifacts, highlighting areas for improvement. In future work, we will explore the integration of 3D imaging and multi-view data to capture volumetric features more effectively, refine preprocessing and augmentation strategies, and develop more robust architectures that ensure biologically meaningful feature attribution. These efforts aim to bridge the gap between predictive modeling and practical application in livestock management.

(a) (b) (c)

Figure 5: Segmented masks generated by the segmentation model. Cow regions are highlighted for rear view (a) and side views (b, c).

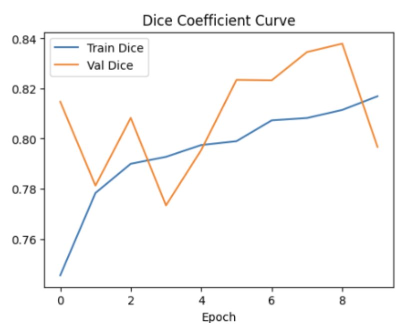
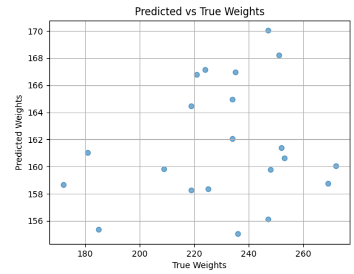


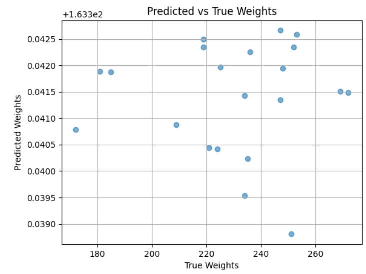
Figure 6: Dice score progression of the segmentation model across training epochs.

Limitations and Future Work

The study is currently confined to regression and segmentation tasks. We only use these two problem settings and do not explore other modeling approaches. Future work expands the scope by incorporating additional architectures and learning frameworks. The dataset consists only of 2D images. The model relies solely on 2D visual inputs, which limits its ability to capture volumetric or structural information. Our future work extends the pipeline to 3D data to capture richer spatial features and improve accuracy.



(a) ReLU Activation



(b) Tanh Activation

Figure 7: Comparison of predicted vs true weights for special test set using different activations.

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